

Chapter 03

State Management and Page Updates

1 lesson • 25 hr

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Chapter 03 • State Management and Page Updates

Practical project of the chapter

15 pages

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State Management and Page Updates

Driving Question

Build a small interactive app using Flet that manages a dynamic list of journal entry titles with add, remove, and favorite toggle functionality, using Python classes for state management.

Production Format

Argumentative essay — 800 to 1500 words.

Context and Provocation

Objective

Build a small interactive app that uses a Python class to manage a list of journal entry titles, allowing you to add, remove, and mark entries as favorites. This exercises dynamic control lists, state encapsulation in classes, and explicit `page.update()`.

Creative Brief

A friend wants a quick tool to manage their daily journal entries. They need to see a list, add new entries, remove old ones, and mark entries as favorites for easy access. They want it to be responsive and feel like a real app. You'll build this component that later can be integrated into the full journal app. You have full freedom on colors and layout.

Objective

Build a small interactive app that uses a Python class to manage a list of journal entry titles, allowing you to add, remove, and mark entries as favorites. This exercises dynamic control lists, state encapsulation in classes, and explicit `page.update()`.

Creative Brief

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Materials and Resources

- Python 3.7 or higher installed
- Flet library installed (pip install flet)
- A text editor or IDE (e.g., VS Code, PyCharm, or even Notepad)
- Terminal or command prompt

Execution Steps

1. Create a new Python file (e.g., `journal_manager.py`) and import flet.
 - Professional insider tip: Start with `import flet as ft` to use shorter aliases.
 - Common mistake to avoid: Forgetting to call `ft.app(target=main)` at the end.
2. Define a Python class `JournalEntry` that holds `'title'` (string) and `'is_favorite'` (bool). Add a method `'build_row(self, page)'` that returns a `'ft.Row'` containing a `'ft.Text'`, a `'ft.IconButton'` for toggling favorite (change icon based on `'is_favorite'`), and a `'ft.IconButton'` for deleting.
 - Professional insider tip: Use `ft.icons.STAR` and `ft.icons.STAR_BORDER` for favorite toggle.
 - Common mistake to avoid: Storing UI controls as instance variables that become stale; always rebuild the row when state changes.
3. Define a class `StateManager` that accepts the `'page'` and initializes an empty list of `'JournalEntry'` objects. Include methods: `'add_entry(title)'` that creates a new entry and calls `'rebuild_ui()'`, `'remove_entry(entry)'` that removes the entry and calls `'rebuild_ui()'`, `'toggle_favorite(entry)'` that flips `'is_favorite'` and calls `'rebuild_ui()'`, and `'rebuild_ui()'` that clears the page column, adds rows for each entry, and calls `'page.update()'`.
 - Professional insider tip: Store `'page'` as an attribute so you can call `'page.update()'` from any method.
 - Common mistake to avoid: Modifying `'page.controls'` directly without clearing the list; use `'page.clean()'` or set `'page.controls = []'`.
4. In the `'main'` function, create a `'ft.TextField'` for new entry title, a `'ft.ElevatedButton'` to add, and a `'ft.Column'` to display entries. Instantiate a `'StateManager'` object, and wire the button's `'on_click'` to call `'state_manager.add_entry(text_field.value)'`. Add the controls to `'page.add()'`.
 - Professional insider tip: Clear the text field after adding to improve UX.
 - Common mistake to avoid: Calling `'page.update()'` inside the event handler instead of through the state manager.
5. Run the app and test: add a few entries, remove one, toggle favorites. Ensure the UI updates correctly without errors.
 - Professional insider tip: Use `'page.update()'` only once per state change in `'rebuild_ui()'` to avoid redundant updates.
 - Common mistake to avoid: Forgetting to call `'rebuild_ui()'` after each mutation; the UI will not reflect changes.

Self-Assessment Checklist

- I can add a new entry by typing a title and clicking the add button.
- The new entry appears in the list immediately.
- I can click the delete button on any entry, and it disappears from the list.
- I can click the favorite toggle button on any entry, and its icon changes between star and star_border.
- The app does not crash when adding, removing, or toggling multiple times.
- The code uses classes to encapsulate state and UI logic (no global variables).
- All state mutations are followed by a single 'page.update()' call.

Bonus Challenge

Add a 'Clear All' button that removes all entries at once. Or, add a simple JSON file persistence: save the list to a file on exit and load it on startup.

Submission Format

Accepted formats:

- text (.txt)

Minimum delivery:

- One .txt file containing the complete Python source code of your app.

Development Guide

1. Research Phase

Look for examples of small apartment designs on sites like ArchDaily or Dezeen. Read news articles on micro-apartments to see both praise and criticism. Note how different designs handle the space-saving vs. comfort tradeoff. Use the course chapters on apartment planning and 3D modeling as your main guides.

2. Organization Phase

Start by stating your main argument — your position on balancing efficiency and comfort. Then outline three to four key points that support your view, using the mandatory references. For each point, plan what evidence or examples you'll use. Also think about one counterargument and how you'll respond to it.

3. Writing Phase

Write an introduction that presents the problem and your thesis. In the body paragraphs, each should focus on one key point — explaining a course concept and applying it to your design example. Use simple language: imagine explaining to a friend. Include specific examples, like how you'd place a bed or choose lighting. End with a conclusion that summarizes your argument and suggests practical tips for designers.

4. Review Phase

Check that your essay clearly states your position. Verify that you've used all four mandatory references correctly. Read it aloud to see if the logic flows smoothly. Ask yourself: would a beginner understand this? Remove any jargon or complex terms without explanation.

Self-Assessment Rubric

1. Class Implementation

- Insufficient — Missing a class or crucial methods; using global variables instead of classes.
- Adequate — Both classes defined, but maybe missing one method or parameter misuse.
- Excellent — Both 'JournalEntry' and 'StateManager' defined correctly; 'StateManager' holds 'page' and 'entries'; methods 'add_entry', 'remove_entry', 'toggle_favorite', 'rebuild_ui' present.

2. UI Row Building

- Insufficient — Row not returned; or controls not added; or favorite icon not dynamic.
- Adequate — Row built but maybe missing delete button or favorite not toggling correctly; uses same row object.
- Excellent — 'build_row' returns a 'ft.Row' with 'ft.Text', 'ft.IconButton' for favorite (icon toggles based on state), and 'ft.IconButton' for delete. No stale controls.

3. State Management

- Insufficient — No 'rebuild_ui' method; or 'page.update()' called in mutation methods causing multiple updates; or column not cleared.
- Adequate — Mutations call 'rebuild_ui' but 'page.update()' may be called elsewhere; column clearing works but maybe with 'clean()' vs 'controls=[]' misused.
- Excellent — State mutations ('add', 'remove', 'toggle') call 'rebuild_ui()'; 'rebuild_ui' clears the column, rebuilds all rows, and calls 'page.update()' exactly once.

4. Main Function & Event Wiring

- Insufficient — No 'main' function; or no event wiring; or missing column; or no 'ft.app' call.
- Adequate — Missing one element (e.g., text field clear) or no 'ft.app' call; but app still works.
- Excellent — 'main' creates 'TextField', 'Button', 'Column'; instantiates 'StateManager(page)'; button 'on_click' calls 'state_manager.add_entry(text_field.value)' and clears text field; all controls added to page; app ends with 'ft.app(target=main)'.

5. Functionality & Robustness

- Insufficient — Crashing on basic operations; entries not appearing; delete/toggle not working.
- Adequate — Basic functionality works but may have minor glitches (e.g., text field not cleared); no crashes.
- Excellent — Adding, removing, toggling work without errors; app handles rapid clicks; no crashes; list updates immediately.

Model Response & Assessment Reference

This guide presents possible paths, not single answers. An excellent essay may defend any of the theses below — or an original thesis not anticipated here — as long as it demonstrates argumentative rigor, correct use of sources, and autonomous critical reflection.

1. Expected Thesis Position(s)

The central question admits multiple legitimate positions. Below are argumentative paths representing robust and distinct directions.

Author note: *The recommended range is 2 to 4 theses. If the assignment has only one defensible answer, keep just Thesis A and adapt the wording. A third thesis often works best as a synthetic or dialectical position.*

1.1. Thesis A — {Short label}

{Full thesis statement, in one or two sentences. It should take a clear position on the central question.}

Main argumentative line: {One paragraph describing how this thesis sustains itself: the central claim, supporting reasoning, implicit framework, and what makes it defensible.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

1.2. Thesis B — {Short label}

{Full thesis statement — a meaningfully different position from Thesis A, not just a softer variant.}

Main argumentative line: {One paragraph describing the central claim, supporting reasoning, implicit framework, and what makes it defensible.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

1.3. Thesis C — {Synthetic/dialectical position, optional}

{Full thesis statement — a meaningfully different position from Thesis A, not just a softer variant.}

Main argumentative line: {Show explicitly how this position produces a new insight that neither A nor B alone can reach.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

2. Key Concepts and Their Correct Use

Author note: Include 3 to 5 key concepts per project. For each concept: precise definition + example of correct application + common pitfall. The pitfall section is critical — it tells the evaluator what to watch for.

2.1. A — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work — not just being named, but actively grounding a claim.}

Common pitfall: {The most frequent way students misuse this concept: over-application, decorative citation, missing nuance, ignoring critiques, or conflating related concepts. Be concrete.}

2.2. B — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work and grounding a claim.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

2.3. C — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work and grounding a claim.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

3. Model Argumentative Structure

The outline below illustrates the structure of an excellent-level essay, regardless of which thesis the student chooses.

Author note: Word ranges are suggestions calibrated for a {total length}-word essay. Adjust proportionally if the assignment is shorter or longer.

3.1. Introduction ({X}–{Y} words)

Contemporary hook: {Describe what kind of opening works for this project — a concrete example, striking data point, scene from culture, or current event. Adapt to the project's domain.}

Bridge to subject: {How the hook should pivot into the project's central question.}

Explicit thesis: The author's position in one or two clear, specific sentences, positioned directly against the central question.

3.2. Development ({X}–{Y} words)

Suggested: 2–3 paragraphs, each with a central argument.

Paragraph 1: {Function and how it advances the thesis.}

Paragraph 2: {Function and how it advances the thesis.}

Paragraph 3: {Function and how it advances the thesis.}

3.3. Counterargument ({X}–{Y} words)

An excellent essay does not merely mention an objection: it presents the strongest counterargument with intellectual honesty and responds consistently.

Author note: *List the best counterargument for each thesis option, with a suggested response. The student should not feel that confronting the objection weakens their position — done well, it strengthens it.*

If thesis is A: {Strongest objection and possible response.}

If thesis is B: {Strongest objection and possible response.}

If thesis is C: {Strongest objection and possible response.}

3.4. Conclusion ({X}–{Y} words)

Restatement of thesis: Reaffirm the position in light of the full argumentation, without repeating the introduction verbatim.

Implications: {What this reflection reveals about something larger — the field, how we think, or a contemporary issue.}

Opening (no new information): A provocative question or forward-looking gesture that broadens the horizon without introducing arguments not developed in the essay.

3.5. Evidence and Data Types to Mobilize

- Specific passages from primary sources.
- Theoretical concepts with citation to the original work.
- Documented contemporary examples: news, official reports, data, or cultural productions.
- References to academic debates covered in the course.

4. Rubric — Excellent Level

Detailed description of what the evaluator should observe in an essay rated “Excellent” on each criterion.

Author note: *The criteria below cover the standard dimensions of argumentative essays. Remove or adapt criteria that do not apply to the specific project.*

4.1. Thesis clarity

What is observed at the EXCELLENT level: The thesis appears in the introduction unequivocally. It is specific and directly engages the project's central tension. The entire essay is organized around it.

4.2. Use of course concepts

What is observed at the EXCELLENT level: Required references are defined precisely before application. Concepts are integrated organically into the argument, and their limits are recognized.

4.3. Quality of argumentation

What is observed at the EXCELLENT level: Each paragraph opens with a clear claim, sustains it with evidence, and connects back to the thesis. Logical progression is clear and the student analyzes rather than merely describes.

4.4. Counterarguments

What is observed at the EXCELLENT level: The student presents the strongest objection to their position and responds with a substantial argument. The thesis emerges stronger from confronting the critique.

4.5. Originality of analysis

What is observed at the EXCELLENT level: The essay goes beyond reproducing course materials. The student offers original connections, asks new questions, or proposes a novel interpretation from familiar evidence.

4.6. Connection to the contemporary

What is observed at the EXCELLENT level: The dialogue between subject and present is specific and grounded. Concrete contemporary examples illuminate both sides.

4.7. Structure and academic norms

What is observed at the EXCELLENT level: Cohesive structure with fluid transitions. Citations follow the established academic pattern, and references are complete and correctly formatted.

5. Common Argumentative Errors

The errors below are the most frequent in this type of project. The evaluator should watch for them.

Author note: Errors 2, 3, and 4 apply to almost any argumentative project. Errors 1 and 5 are project-specific slots where particular risks may be described.

5.1. {Project-specific error}

How it appears in the text: {Verbatim-style example showing the error in action.}

How to correct: {Concrete, actionable advice on what to do differently.}

5.2. Strawman fallacy

How it appears in the text: The student presents the counterargument in its weakest, easiest-to-refute form instead of confronting the strongest version.

How to correct: Reformulate the counterargument as an intelligent interlocutor would present it, then respond to that reason.

5.3. Decorative use of concepts

How it appears in the text: The student names a concept but never applies it — it appears as a loose citation, disconnected from the argument.

How to correct: After citing a concept, explain exactly how it grounds the specific argument of that paragraph.

5.4. False equivalence

How it appears in the text: The student treats as equivalent two phenomena that operate in radically different contexts or functions.

How to correct: Make differences explicit before drawing parallels. Use precise comparative language. Comparison is not equation.

5.5. {Project-specific error}

How it appears in the text: {Example}

How to correct: {Correction}

Final reminder: *The quality of reasoning is always more important than the position defended. An essay that defends an unpopular thesis with rigor outperforms one that defends a consensus thesis superficially.*

